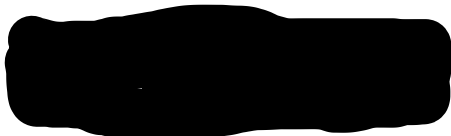


**RF POWER
MEASSUREMENT KIT
(0.2T, 0.5T, 0.7T, 1T, 1.5T)
S/N CHI006770**



IMPORTANT NOTE:

Please use this as a reference guide. Refer to the appropriate GE Service Document to fully adjust or check your RF Amplifier.

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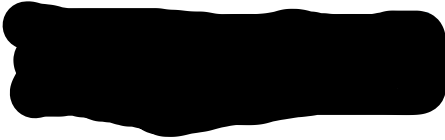


1. SYSTEM CONFIGURATION

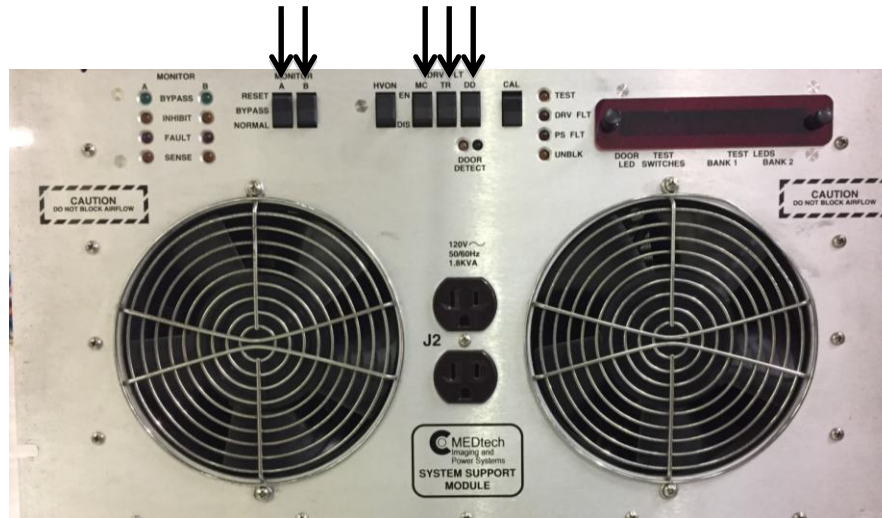
Disable TR, DD and MC



OR

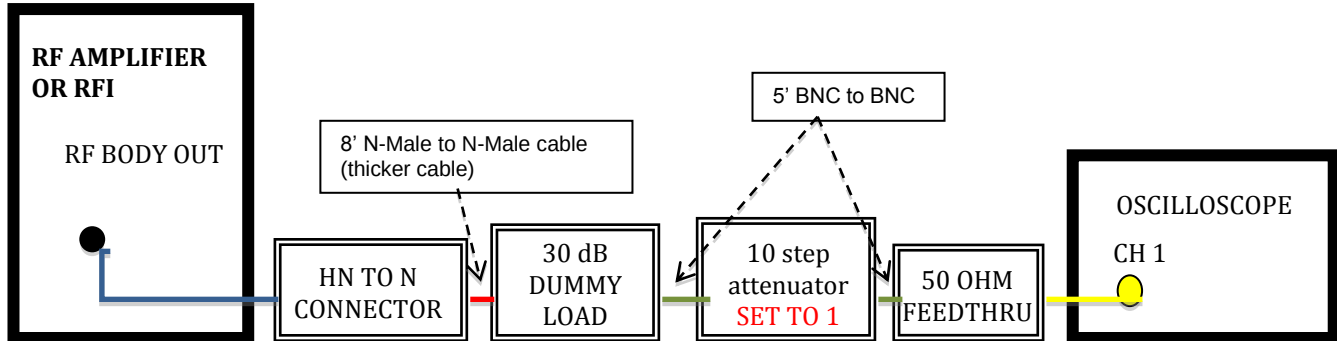


**Disable TR, DD and MC and
set Monitor A and B
in bypass mode.**



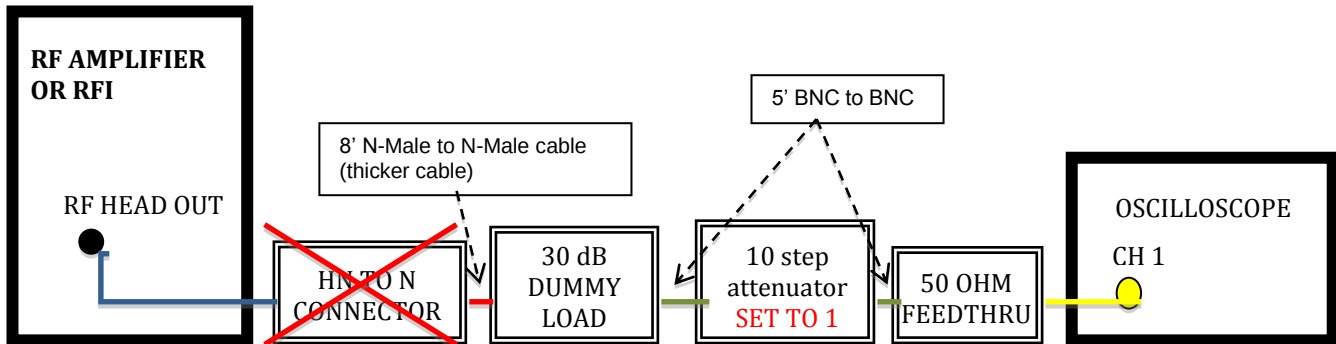


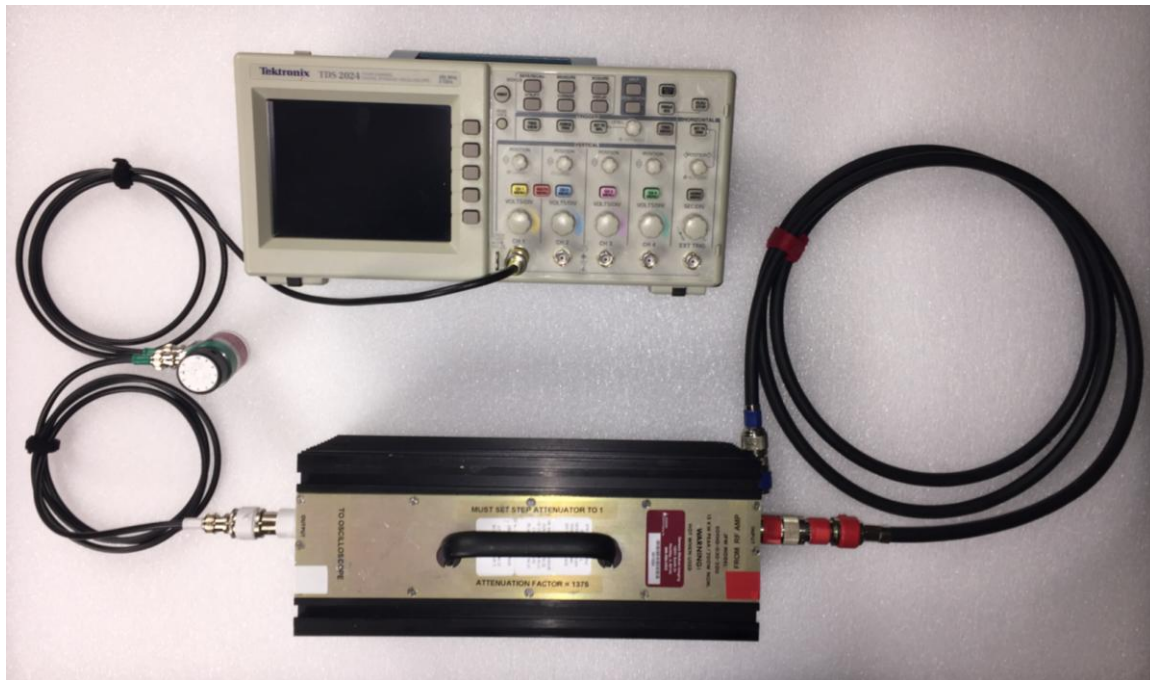
2. BODY RF POWER ADJUSTMENT CONFIGURATION





3. HEAD RF POWER ADJUSTMENT CONFIGURATION







4. PROTOCOL SETUP

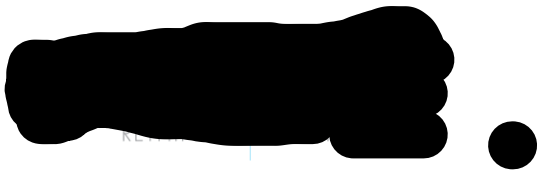
BODY (14x and above):

1. Start a New patient:
 - a. Patient ID: **geservice**
 - b. Name: **rf body cal**
 - c. Weight: **300**
2. Make sure Service protocols are selected.
3. Select Other.
4. Click on **1.5T apb cal** and select series 1 for body.
5. Landmark in the body area with no coils or phantoms.
6. Send to scan.
7. Save Series.
8. Under Research Operations, select Display CVs.
9. Change Calmode to 5. (Make sure to hit Enter after editing the value)
10. Accept.
11. Research operations.
12. Download.
13. Manual prescan and increase the TG slowly to 200 while monitoring your waveform in the scope.



Head (14x and above):

1. New Series
2. Make sure Service protocols are selected.
3. Select Other.
4. Click on **1.5T apb cal** and select series 2 for head.
5. Landmark in the head area with no coils or phantoms on the table.
6. Send to scan.
7. Save Series
8. Under Research Operations, select Display CVs.
9. Change Calmode to 5 (Make sure to hit Enter after editing the value)
10. Accept.
11. Research operations.
12. Download.
13. Manual prescan and increase the TG slowly to 200 while monitoring your waveform in the scope.



BODY (12x and prior)

PATIENT INFORMATION

Patient Id **geservice**

Patient Name **body rf**

Weight (Lb) **300** IMPORTANT

PATIENT PROTOCOLS [Patient Position]

PATIENT POSITION

Patient Position [>][**Supine**]

Patient Entry [>][**Head First**]

Coil [...][**Body**][Accept]

IMAGING PARAMETERS

Plane [>][**Axial**]

Mode [>][**2D**]

Pulse Seq [...][**Spin Echo**]

[Accept]

Imaging Options **none** (default)

Psd Name **cal**

Protocol no entry

SCAN TIMING

of Echoes 1 (default)

TE **25**

TR **55**

SCANNING RANGE

FOV **24**

Slice Thickness **5**

Spacing **0**

Start **0**

End **0**

Slices **1** (default)

L/R Center **0** (default)

P/A Center **0** (default)

Table Delta **0.00** (default)

ACQUISITION TIMING

Freq **256**

Phase **128**

NEX **2**

Freq Dir [>][**A/P**]

[Save Series]

[Research Operations][Display CVs]

calmode **5**

[Accept]

[Research Operations][Download]

REF: [REDACTED]



HEAD (12x and prior)

[New Series]

PATIENT POSITION

Patient Position [>][**Supine**]

Patient Entry [>][**Head First**]

Coil [...][**Head**][Accept]

IMAGING PARAMETERS

Plane [>][**Axial**]

Mode [>][**2D**]

Pulse Seq [...][**Spin Echo**]

[Accept]

Imaging Options **none** (default)

Psd Name **cal**

Protocol no entry

SCAN TIMING

of Echoes 1 (default)

TE **25**

TR **55**

SCANNING RANGE

FOV **24**

Slice Thickness **5**

Spacing **0**

Start **0**

End **0**

Slices **1** (default)

L/R Center **0** (default)

P/A Center **0** (default)

Table Delta **0.00** (default)

ACQUISITION TIMING

Freq **256**

Phase **128**

NEX **2**

Freq Dir [>][**A/P**]

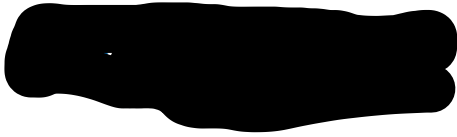
[Save Series]

[Research Operations][Display CVs]

calmode **5**

[Accept]

[Research Operations][Download]



5. RF POWER CALCULATIONS

To calculate the power needed use the following formula:

$$\frac{(V_{\text{peak}})^2}{100} \times \text{dummy load and cable attenuation factor} = \text{RF Power}$$

The dummy load and cable attenuation factor for this kit is:

1375

Typical values:

Others:

16000 Watts (1.5T Body) = **34.1Vp**

3000 Watts = 14.8Vp

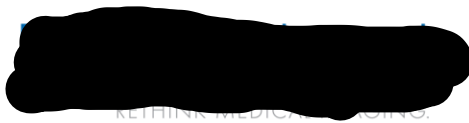
4000 Watts = 17.1Vp

2000 Watts (1.5T Head) = **12.1Vp**

6000 Watts = 20.9Vp

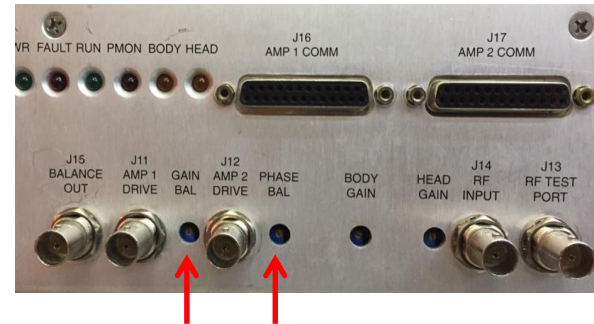
8000 Watts = 24.1Vp

ALWAYS VOLTS PEAK!!



6. RFI CALIBRATION

RFI requires calibration of Gain Balance and Phase Balance (these are 30-turn potentiometers). If during Body Gain adjustments the system keeps tripping, turn Gain Balance pot to midrange and the Phase Balance pot to minimum (all the way CCW). Try adjusting Body Gain again and if it still trips, adjust Gain Balance and Phase Balance with TG at 170 before adjusting Body Gain. Recheck Calibrations after Body Gain is adjusted at 200 TG.



Gain Balance and Phase Balance Calibration

At the front of the RFI adjust the Phase Balance pot fully clockwise (CW).

To adjust Gain Balance, zoom in in the top portion of the waveform and adjust the Gain Balance pot to minimize the waveform.

To adjust Phase Balance, zoom in on the top portion of the waveform and adjust the Phase Balance pot counter-clockwise (CCW) to maximize the waveform.

Check Body adjustment when done.

7. Oscilloscope



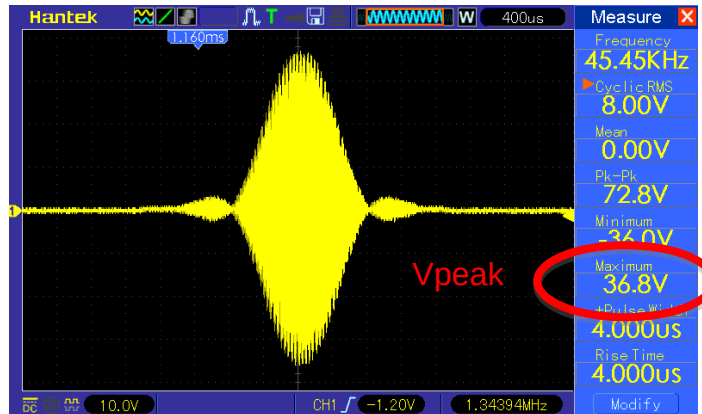
HANTEK:

There are three different setups stored in the scope:

Setup 0: Body mode 10V/div and 400 μ s/div (probably most accurate reading)

Setup 1: Body mode 5V/div and 400 μ s/div (Zooms in on the top part of the waveform)

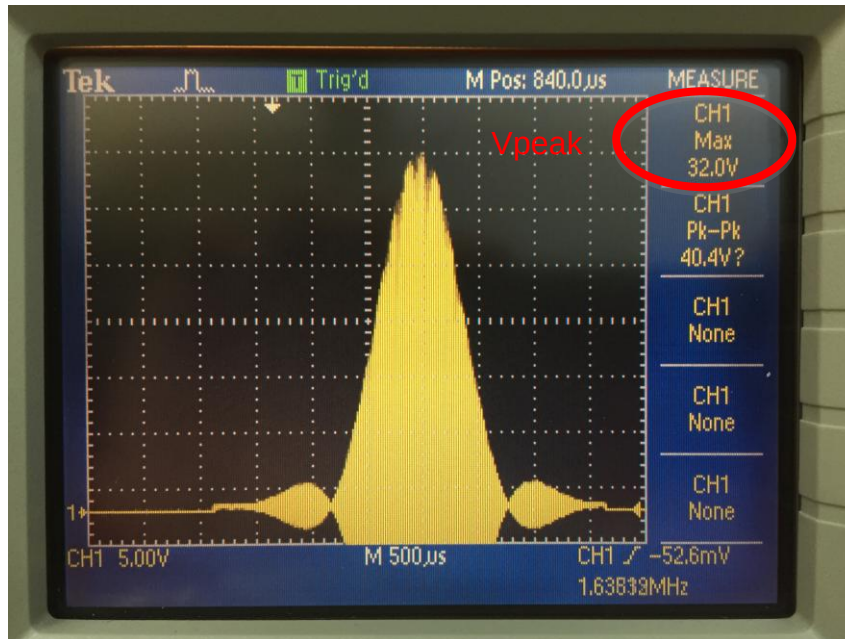
Setup 3: Head mode 2V/div and 400 μ s/div





Tektronix:

Set up to 5V/div and around 840 μ s/div for body and 600 μ s/div for head, gives you a good resolution.





8. Packaging Information

Part #	Description	Quantity
PE3067-60-00	Cable 5' BNC to BNC RG58 50 Ohm	2
PE3062-96-00	Cable 8' N-MALE to N-MALE RG214/U	2
PE9194-00	HN MALE to N FEMALE	2
PE6008-50-00	BNC Feedthru 50 Ohm	2
PE9002-00	BNC FEMALE to N MALE	2
PE9003-00	BNC "T" FEMALE-FEMALE-FEMALE	2
PE9006-00	N FEMALE to N FEMALE	2
PE9084-00	BNC FEMALE to BNC FEMALE	2
PE7034-3-00	10 STEP ATTENUATOR	1
	BNC 50 OHM TERMINATOR	1
	BNC DC BLOCK	1
	RF LOOP (OPTIONAL)	1
	30 DB, 200W, 50 OHM DUMMY LOAD	1
	USB Memory	
	200 MHz OSCILLOSCOPE	1



